SIH26173 iTantra: research, gap analysis and build plan

Sep 28, 2026 · @ire homes

The STT → text → TTS pipeline over Wi-Fi Direct or Bluetooth is already built by many teams, so it cannot be your innovation; the defensible work is measured Indian-language accuracy under noise, a script-efficient wire format, uncertainty-aware delivery, and a link emulator that proves the value on genuinely narrow links.

Bottom line

Do not pitch "STT + TTS over Wi-Fi Direct" as the innovation. More than ten public repositories already implement it for this PS, at least three report two-phone tests, and an open-source Meshtastic plugin already does speech-to-text over LoRa with speech output on the receiver (English only). Pitch a measured system built on that baseline.

Five findings that drive everything below

1. **The baseline is crowded.** Team repos report IndicConformer for STT, VITS/Piper for TTS, Protobuf packets of roughly 50–300 bytes, and Bluetooth RFCOMM or Wi-Fi Direct. Treat this as table stakes, not novelty.
2. **Text is tiny, but the edge over a good codec is 4–9×, not 1000×.** On my emergency-phrase test corpus a 10-second utterance costs 116–238 bytes as script-aware text, against 875 bytes for Codec2 700C and 7,500 bytes for Opus at 6 kbps. Only against raw PCM does the ratio pass 1,000×.
3. **Wi-Fi Direct and Bluetooth are not narrow links.** The saving matters when goodput is near 1 kbps (LoRa, satellite short messages, HF). I computed that LoRa LongFast needs about 1.01 s of airtime per second of Codec2 700C voice, so voice cannot stream, while a 10-second Hindi utterance as text fits in one packet. You need a link emulator to show this on phones.
4. **UTF-8 wastes 2.4–2.8× on Indic scripts** (measured). Script-aware coding is a cheap, defensible gain and decides whether a 10-second Tamil utterance fits in one LoRa packet.
5. **The dangerous failure is silent.** The listener hears a fluent but wrong sentence. I found no public implementation that tells the receiver how uncertain the recognition was. That is where the accuracy criterion (reported as 40% of the score) can be won.

Recommended solution: a verified semantic voice link. The baseline pipeline plus three contributions, each with a number you can defend:

- **Indic-efficient wire format:** script-aware bytes, a short header, an optional phrase codebook for alert traffic.
- **Uncertainty-aware speech path:** sender read-back before sending, per-word confidence, a critical-token guard (numbers, places, medical words), and receiver-side marking of unclear words.
- **Reliable delivery plus a constrained-link emulator:** ACK and retry, de-duplication, priority, store-and-forward, and sweeps over bandwidth, loss and disconnection.

What to build first: Hindi and English end to end with instrumented latency, then the wire format, then confidence and read-back, then the emulator and benchmarks, then more languages. Sections 2 and 8 explain why, and section 16 lists what you must not claim.

Evidence quality. I could not open the official SIH portal text. Requirements marked "reported" come from team repositories that quote it, and must be checked against your portal copy (section 2 has the checklist). Bandwidth numbers are my own measurements on a small corpus, labelled as such.

1. Official PS: what is verified and what is only reported

Only the ID, title, organisation and category are confirmed by several independent listings; every detailed requirement below is second-hand from team repositories and must be checked against the text on your SIH portal.

Verified from listings (SIH 2026 problem-statement browser, Zaid Sayyed's PS index, tanmayjoshi01/iTANTRA)

Field	Value
PS number	SIH26173
Title	iTantra – Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
Organisation	Indian Space Research Organisation (ISRO), Department of Space
Category	Software

Field	Value
Theme	Listings disagree: "Smart Automation" on two, "Miscellaneous" on others; check the portal

Reported requirements (not verified against the portal)

Requirement	Reported by	Confidence
Scoring: Efficiency 20%, Accuracy 40%, Latency 20% (the fourth 20% is not stated anywhere I found)	Precise-Goals audit, abdulsamad1366	Two repos agree; verify
Ten languages: Hindi, Gujarati, Marathi, Kannada, Malayalam, Tamil, Telugu, Odia, Bengali, English	anubrr21, abdulsamad1366, Precise-Goals ("10 mandated languages")	Three repos agree; verify
Transport is Wi-Fi and Bluetooth only	Precise-Goals audit	One repo; verify
Push-to-talk walkie-talkie mode, plus a phone-like mode	Precise-Goals audit	One repo; verify
Alerts announced at highest volume, non-interruptible	Niranjan266 (quotes the PS)	One repo; verify
Four latency figures are required (which four is not stated)	Precise-Goals audit	One repo; verify
Background: voice streaming needs above 64 kbps; text saves bandwidth but voice serves people regardless of literacy	abdulsamad1366	One repo; verify

No source says translation between languages is required. "Multilingual" appears to mean speech recognition and speech output in each language.

Requirement split

Class	Items
A. Explicit (reported, verify)	On-device STT and TTS; ten languages; text over Wi-Fi/Bluetooth; push-to-talk and phone-like modes; non-interruptible alerts; the three scoring criteria
B. Inferred by teams, not by the PS	Protobuf packets, zlib, Silero VAD, BLE broadcast mesh, hop relay, AFSK, DTN
C. Optional	Translation, prosody, phrase codebook, mesh, store-and-forward, backend, LoRa bridge
D. Do not claim unless implemented and measured	Accuracy in all ten languages, full offline operation, mesh, range, battery life, real-disaster performance

Check these on the portal before you freeze scope

- [ ] Exact Background, Description and Expected Outcome wording
- [ ] Whether the ten languages are named, and whether all ten are mandatory for the demo
- [ ] Whether translation is required or optional
- [ ] Which transports are allowed (Wi-Fi, Bluetooth, anything else)
- [ ] What "non-interruptible alert" must do on Android (volume, Do Not Disturb)
- [ ] The fourth scoring criterion and the four latency figures
- [ ] Deliverables (APK, video, report) and any hardware assumptions

If you paste the portal text, I can turn it into a line-by-line compliance matrix.

2. Existing systems, products and open-source projects

No single product or project solves SIH26173 end to end for ten Indian languages, but every piece exists somewhere, and the pipeline itself has been built repeatedly by other teams. Types are labelled so a prototype is never read as a product.

System	Type	What it actually does	Solves SIH26173?
Google Translate Live, offline mode	Commercial product	On-device conversation translation, only on Pixel 9, 10 and 11; 11 languages, every pair includes English; Hindi is the only Indian language (BGR)	Partly: on-device STT, MT and TTS on one phone. No device-to-device transport, one of ten languages, one phone family
Samsung Interpreter	Commercial product	Galaxy AI offline packs; a third-party guide lists Hindi among them (unverified by Samsung), needs Galaxy S24 or newer	Same gap as above; not checked against Samsung's own docs
Bhashini	Government platform (hosted APIs)	ASR, translation, TTS for 22 languages as cloud services; a 2024 open ticket asks for an offline Hindi ASR under 150 MB at about 80% accuracy	No: cloud-first; offline was still a request in 2024
AI4Bharat models	Open-source models (IIT Madras lab)	IndicConformer ASR, Indic-TTS, IndicTrans2, IndicVoices data	Components only, no communication layer
Meshtastic with its ATAK plugin	Open-source project, community-deployed	LoRa mesh; the plugin records speech, converts it to text with Vosk, sends it over the mesh and speaks it on receivers with TTS enabled; English only	Closest transport prior art; no Indian languages
Briar	Open-source app, security-audited in 2017 per reviews	Syncs over Bluetooth, Wi-Fi, memory cards or Tor; mailbox for offline receivers	Delivery-layer prior art; no speech

System	Type	What it actually does	Solves SIH26173?
Bridgefy, BitChat, OSHI	Commercial and open-source apps	Bluetooth and Wi-Fi mesh text chat	No speech; no Indian-language voice
Codec 2 / FreeDV	Open-source codec	Voice at 700–3200 bit/s for HF and VHF radio; its maintainers say the 700C mode is not actively maintained	The baseline you must beat on narrow links
Lyra V2	Open-source codec	3.2, 6 and 9.2 kbps neural voice codec, TensorFlow Lite, tested on Android	Second baseline for compressed voice
STCTS	Research preprint	STT, text compression and TTS at about 80 bps with sparse prosody and speaker embedding; evaluated on English LibriSpeech	Same idea, not Indic, not shown on phones
Other SIH26173 team repos	Student prototypes, unreviewed	See list below	Same baseline; claims unverified
uktam and offline-translator	Open-source Android apps	IndicConformer through sherpa-onnx plus Sarvam Translate through llama.cpp with RAM-based model choice; Firefox translation models	Shows on-device translation is feasible on phones

Other teams on this PS (all unreviewed, claims not verified by me)

- Precise-Goals: two physical devices, Wi-Fi Direct and Bluetooth RFCOMM, IndicConformer plus VITS through sherpa-onnx, Protobuf packets of 50–300 bytes. Its own audit lists TTS for only five of ten languages and reports a compulsory model bundle near 2.18 GB.
- Niranjan266: BLE broadcast mesh, an 11-byte frozen header, a phrase codebook, English and Tamil packs, Bluetooth test between two handsets, 226 unit tests.

- psinghal01 and vikranthsai310: identical write-ups claiming ten languages, relay, alerts and an airplane-mode demo video.
- abdulsamad1366: Flutter, sherpa-onnx, zlib plus a 23-byte header; claims ten languages and a latency table without shown measurements.
- psb-001 evidence repo: FLEURS-based error rates and a noise sweep; states that Gujarati, Kannada and Odia have no speech output.
- rushi-m36: Android's own recognizer and TTS over TCP on a local network.

Do modern phones already give you the pieces?

Capability	Status
Offline speech recognition	Yes for some languages on some devices; Indian-language coverage varies by phone and Android build. Test on your target phones
Offline translation	Pixel 9–11 only, Hindi is the only Indian language there
Offline TTS	Piper/VITS voices run on CPU through sherpa-onnx; system TTS voices vary by device
Peer-to-peer link	Yes: Nearby Connections covers about 100 m over Bluetooth and Wi-Fi

So the pieces exist. What does not exist publicly is a measured, ten-language, phone-tested chain with delivery guarantees and a way to show its value on a narrow link. That is where your work has to sit.

3. Research papers

The literature supports the idea (speech can be rebuilt from very small semantic representations) but none of it targets Indian languages on phones, so it strengthens your framing without replacing your engineering. I read abstracts and excerpts of these, not full papers; open each one before citing it on a slide.

Paper	What it shows	Adapt	Do not claim
Polyak et al., Interspeech 2021, speech resynthesis from discrete self-supervised units (Meta AI and Hebrew University authors)	A 365 bps codec from HuBERT units, pitch and speaker codes rated better than Opus 9 kbps, Codec2 2.4 kbps and LPCNet 1.6 kbps on English VCTK	Use as the "semantic codec" comparison point for your text bitrate	Not Indic, needs HuBERT and a GAN vocoder, not shown on phones
STCTS, arXiv 2512.00451, Nov 2025	Text about 70 bps plus sparse prosody under 14 bps gives about 80 bps; reported 75× below Opus 6 kbps on LibriSpeech; finds prosody sampled sparsely or densely works, mid-rates do not	Sparse prosody: one byte for pace and urgency; amortised speaker data; the same comparison logic	Preprint, English, server-class TTS; its MOS is not your MOS
Collette, Greenwood, Nicoll, arXiv 2509.15462, Sep 2025	Semantic compression matches codecs on transcription, sentiment and speaker verification at 2–4× lower bitrate	Its own limitations list (overlapping speakers, one-time timbre errors) doubles as your limitations slide	Not a phone system
Leschanowsky et al., Interspeech 2025, SITool benchmark	Subjective intelligibility of AMR, AMR-WB, EVS, Codec2 (0.7 and 2.4 kbps), LPCNet, Lyra V2, Mimi, SemantiCodec	Method for a small listening test of your TTS output	The excerpt hints at an intelligibility drop for neural codecs below about 1.1 kbps; read the paper before quoting it

Paper	What it shows	Adapt	Do not claim
LMCodec2, ScienceDirect 2024	A sub-1 kbps voice codec aimed at Beidou-3 short messages; frames satellite messaging as a real narrow link	Cite as a real narrow-link use case	Details not verified beyond the abstract
Javed et al., ACL Findings 2024, IndicVoices	7,348 hours, 22 languages, 16,237 speakers; 9% read, 74% extempore, 17% conversational; 1,639 hours transcribed at publication	Source for spontaneous-speech test data	Its test set is not public (BhasaAnuvaad)
BhasaAnuvaad, arXiv 2411.04699	Existing models do well on read speech and clearly worse on spontaneous speech	Justifies testing on spontaneous and noisy speech, not only read sentences	Concerns speech translation models, not only ASR
Sankar et al., NeurIPS 2024, IndicVoices-R	1,704 hours of TTS-grade speech, 10,496 speakers, 22 languages	Only if you fine-tune a missing voice (red for a student team)	Do not promise new voices
Indic-TTS, AI4Bharat (ICASSP 2023 per the repo)	FastPitch plus HiFi-GAN V1 worked best across 13 Indian languages	Quality reference	Heavier than Piper/VITS for phones
Rasmalai / Indic Parler-TTS, arXiv 2505.18609	13,000 hours over 24 languages; trained on 32 H100 GPUs	None for phones	Server-class; do not pitch it as on-device
IndicTrans2, TMLR 2023 (Gala, AI4Bharat et al.)	Translation for all 22 languages; 1.1B-parameter base, about 211M distilled	Optional translation stretch	Phone speed is unproven

Paper	What it shows	Adapt	Do not claim
On-device streaming ASR, arXiv 2604.14493	Post-training quantization cut an English model from 2.47 GB to 0.67 GB within 1% absolute WER	Supports quantizing Indic models	English, different model; re-measure yours
SN-WER, arXiv 2606.02548	A script-normalized error rate for multi-script Indic ASR	Report CER and WER, and note script effects	Companion metric, not a replacement
RFC 9171, IETF 2022 (Burleigh, Fall, Birrane)	Bundle Protocol 7, a store-carry-forward overlay for intermittent links	Borrow ideas: lifetime, custody, opportunistic forwarding	Do not claim BPv7 compliance

What the papers do not give you: any measured Indian-language STT to TTS chain on a phone. Your benchmark section fills that.

4. Indian-language AI stack: ASR, TTS and translation

Per-language IndicConformer (int8, about 140 MB each) and Piper/VITS voices (about 21 MB each in int8) are the only combination I found that plausibly runs offline on a mid-range phone for most of your ten languages; the 600M multilingual model and the large TTS models do not fit that budget. Nobody has published on-phone accuracy or speed for this stack, so that measurement is part of your contribution.

Speech recognition

Model	Languages	Size and runtime evidence	Licence	Verdict
IndicConformer, per language (hybrid CTC-RNNT)	One model per language, all 22 have cards	Encoder about 120M parameters, 17 Conformer blocks, dimension 512. A third-party int8 CTC export for gu, pa, bn, mr, ml, te, ta, kn is 140,337,395 bytes each and loads through sherpa-onnx	MIT on the cards; one team repo says Apache-2.0, so check	Best phone candidate; needs your own speed, RAM and accuracy measurement
IndicConformer-600M multilingual	22	600M parameters. A third-party int8 build shrinks disk from about 2.4 GB to 0.74 GB and RAM from 3.34 GB to 2.50 GB, CPU real-time factor about 0.29 (hardware not stated)	MIT	Edge laptop only; too heavy for low-end phones
AI4Bharat 30M IndicConformer	Several	Listed as deployable on Android over websockets; older model	Check	Look at it for very low-end phones; availability unverified

Model	Languages	Size and runtime evidence	Licence	Verdict
Whisper	Many	One team reports about 100% word error on a single Tamil test sentence (Niranjan266)	MIT	One synthetic sentence is weak evidence; measure it yourself before dismissing it
Bhashini ASR	22	Hosted API	Platform terms	Cloud only; excluded by the offline requirement

One team repo (psb-001) reports that a character-level recognizer with no language argument emitted characters from other scripts, and that pruning the vocabulary to the target script fixed it. Treat that as a cheap accuracy trick to test.

Speech output

Model	Languages and size	Licence	Verdict
Piper voices (VITS)	Hindi, Telugu, Bengali, Marathi, Malayalam, Nepali, Urdu, English; about 60–77 MB in fp32; int8 builds about 21 MB for Hindi, Malayalam and English (Precise-Goals audit)	Repo is MIT; each voice's dataset licence differs (a release note lists Marathi CC-BY-SA 4.0, Telugu CC-BY 4.0, Bengali CC-BY-SA 4.0); some voices trace to a base with a commercial restriction (sherpa-onnx #3947)	Default choice; record each voice's licence

Model	Languages and size	Licence	Verdict
Meta MMS-TTS (VITS)	Tamil, Kannada, Odia, Marathi, Telugu and more; about 109 MB, 36.8 MB after quantization in one team's build	CC-BY-NC 4.0, non-commercial (release note)	Fine for a competition, must be declared; a team audit found the sherpa-onnx release only ships an English MMS voice, so you convert it yourself
AI4Bharat VITS (vits_rasa_13)	13 languages by name	Listed MIT; check the dataset licence	Promising for coverage; verify Android export
Indic Parler-TTS	21 languages	MIT listed	Server-class; not for phones
Kokoro-82M	Includes Hindi	GPL app wrapper	One app reports 2–3 minutes per minute of audio on budget phones (VoxSherpa release); not real time

Team reports conflict on coverage: one says Gujarati, Kannada and Odia have no speech output, another claims Indic-VITS covers all ten. Resolve this yourself in week one by synthesizing one sentence per language on a phone.

Translation (optional)

IndicTrans2 distilled models are about 200M (English to Indic) and 320M (Indic to Indic) parameters. uktam shows an Indian-language STT plus llama.cpp translation app choosing model size by phone RAM. Nothing here gives you a measured translation speed on your phones, so keep translation out of the core.

What can run where

Where	What
Fully on a mid-range phone (to be measured)	Silero VAD (under 2 MB per one team), per-language IndicConformer int8 CTC, Piper int8 voices

Where	What
On the phone with a caveat	MMS voices (non-commercial), IndicTrans2 distilled (speed unknown)
Edge laptop as a base station	600M multilingual ASR, large TTS, 1.1B translation
Cloud	Bhashini and similar APIs; keep out of the core path

Before you trust any model, measure on the phone: cold load time, peak memory (not just Java heap), real-time factor, thermal drift over ten minutes, and error rate on the phone against the same audio on a desktop (quantization drift).

5. Communication technologies

Every short-range option carries far more than a text message needs, so the choice between them is about setup friction, reliability and range, not bandwidth; build one transport interface, ship Bluetooth RFCOMM and Wi-Fi Direct behind it, and wrap both in a link emulator.

Technology	What is known	Good for	Watch out for
Bluetooth Classic (RFCOMM socket)	A team repo reports a 62-byte message delivered between two different phone brands (Niranjan266); reviews of Briar quote 10–30 m range	Simplest reliable socket for a two-phone demo	Pairing prompt, OEM differences, shorter range
Wi-Fi Direct (Wi-Fi P2P)	Available since Android 4.0; Google says it reaches much further than Bluetooth (Nearby docs)	The transport the PS names; groups of devices	Group-owner negotiation, connection prompts, OEM behaviour, needs the Nearby-devices permission on Android 13+

Technology	What is known	Good for	Watch out for
Bluetooth LE	Small packets; no pairing needed for advertising	Discovery, presence beacons, tiny alert broadcasts	Low throughput, small MTU, background limits
Nearby Connections	About 100 m, chooses Bluetooth or Wi-Fi itself; strategies P2P_CLUSTER (many-to-many, default), P2P_STAR, P2P_POINT_TO_POINT; Android 12+ needs BLUETOOTH_SCAN and BLUETOOTH_CONNECT, Android 13+ NEARBY_WIFI_DEVICES	A working fallback and a way to share model files between phones	Needs Google Play services; you lose control of link details
Multi-hop mesh	Not provided by any of the above; Airpass shows connect-sync-drop epidemic routing over Nearby	Extending range past one hop	You must write flooding, TTL and de-duplication yourself
Delay-tolerant networking	RFC 9171 defines store-carry-forward for intermittent links; Briar syncs through intermediaries and a mailbox	Ideas for the outbox: lifetime, retry, custody	Full Bundle Protocol is out of scope for a student build
LoRa via Meshtastic	Payload about 237 bytes; typical range 2–5 km (NodakMesh); third-party Android apps talk to it through its AIDL service and can use a private port number in 256–511	The real narrow link; the honest demo of why text beats audio	Needs two LoRa boards; bandwidth about 1 kbps; radio regulations

Recommendation. Make Bluetooth RFCOMM the first demo transport (least friction), Wi-Fi Direct the second (the PS names it), and treat Nearby Connections as a fallback. Put a **link emulator** (rate limit, loss, jitter, forced disconnects) under all of them, because your phones' links are megabits wide and cannot show the narrow-link advantage on their own. Treat the Meshtastic LoRa bridge as a yellow stretch goal.

Reported rather than verified: one team says Android needs the INTERNET permission for any socket, including a purely local Wi-Fi one, while BLE does not. Check this before you claim "no internet permission".

6. Does sending text really save bandwidth?

Yes, but the honest headline is "73–87% smaller than the best low-rate voice codec and 94–99% smaller than 3–6 kbps voice codecs", not "98% smaller than audio"; and on Wi-Fi or Bluetooth the saving does not matter until you emulate a narrow link. These figures are my own measurements and calculations, run this session, on a small hand-written corpus, so replace them with numbers from real recognizer output before they go on a slide.

Method. Eight emergency-style sentences in each of Hindi, Marathi, Tamil, Telugu, Bengali and English (flood, injury, water, road, medicine, location). Sizes measured in Python for UTF-8, a script-aware one-byte mapping (ASCII stays, each Indic Unicode block character becomes one byte), raw deflate, Brotli, and deflate with a preset dictionary built from the other seven sentences (optimistic, because the sentences share phrases). Speech rate assumed 2.5 words per second for every language, which probably overstates Tamil and Telugu because their words are longer.

Bytes per sentence (mean of 8)

Language	Characters	UTF-8	Script-aware 1 byte	Deflate on 1 byte	Preset dictionary (optimistic)
Hindi	45	117	45	49	36
Marathi	46	120	46	49	37
Tamil	60	164	60	62	41
Telugu	49	132	49	54	41
Bengali	44	118	44	49	35
English	48	48	48	49	34

UTF-8 costs 2.4–2.8× more than one byte per character on Indic scripts. Generic compression on a 45-byte message gains nothing; a shared dictionary does.

The same 10 seconds of speech in different representations

Representation	Bytes	Note
Raw PCM, 16 kHz, 16-bit	320,000	256 kbps

Representation	Bytes	Note
G.711 μ-law	80,000	64 kbps
Opus, 6 kbps	7,500	Opus minimum
AMR-NB, 4.75 kbps	5,938	Cellular voice
Lyra V2 or Codec2 3200	4,000	3.2 kbps
Codec2 700C	875	Lowest classic mode
Text, Tamil, 1 byte per char	about 238	25 words; likely an overestimate
Text, Hindi, 1 byte per char	about 116	25 words
Text, Hindi, UTF-8	about 300	What a naive app sends
HuBERT-unit codec, research	about 456	365 bps
STCTS, research	about 100	80 bps, English

Against Codec2 700C the Hindi text is 7.5× smaller and the Tamil text 3.7× smaller (87% and 73% less). Against Opus 6 kbps it is 96.8–98.5% smaller. The "98.4%" figure some teams quote only holds against 64 kbps or PCM audio.

Framing matters as much as compression. With a 12-byte application header and 20 bytes for an authentication tag and counter, a Hindi sentence becomes about 77 bytes on the wire, in line with the 50–300 bytes other teams report. Header design is where a 45-byte payload is won or lost.

Where it becomes decisive: a narrow link. I computed LoRa time-on-air with the standard Semtech formula (explicit header, CRC on, 16-symbol preamble, coding rate 4/5). Check it with a LoRa airtime calculator before quoting.

LoRa preset	Nominal bit rate	100-byte packet	237-byte packet	1 s of Codec2 700C voice (104-byte packet)
LongFast (SF11, 250 kHz)	1,074 bps	1.01 s	2.03 s	1.01 s of airtime: cannot stream
MediumFast (SF9, 250 kHz)	3,516 bps	0.29 s	0.60 s	0.30 s: possible, uses a third of the channel

LoRa preset	Nominal bit rate	100-byte packet	237-byte packet	1 s of Codec2 700C voice (104-byte packet)
ShortFast (SF7, 250 kHz)	10,938 bps	0.09 s	0.19 s	0.09 s

On LongFast a 10-second Hindi utterance as script-aware text is one packet (about 148 bytes with headers, roughly 1.4 s of airtime). The same Tamil utterance in UTF-8 needs four packets. That is the argument for text over voice, and for script-aware coding.

Trade-offs to state openly

- **Latency:** text leaves after the speaker stops, so delay is the utterance plus recognition, link and synthesis time; a codec streams as it goes. Measure it.
- **Compute and battery:** recognition and synthesis cost far more CPU than a codec.
- **What is lost:** the speaker's voice, emotion, background sounds, and any word the recognizer gets wrong.
- **Where codecs are enough:** at 3 kbps and above (Wi-Fi, Bluetooth, most cellular) a codec is fine; the text advantage is about narrow, lossy or intermittent links, where a small message can be retransmitted and stored.

7. Fragmentation, verified pain points and the real gap

The fragmentation is real but narrower than "existing apps don't work offline": products that have speech intelligence lack a device-to-device link, products that have a link lack speech, and the few that combine them do not cover Indian languages or measure accuracy under noise.

Who covers which layer

System	Speech on device	Indian languages	Device-to-device link	Store-and-forward	Works below 1 kbps
Google Live Translate, offline	Yes, Pixel 9–11 only	Hindi only	No	No	No
Bhashini	Cloud	22, hosted	No	No	No
AI4Bharat models	Components	22 (ASR), fewer (TTS)	No	No	No
Meshtastic with ATAK plugin	Yes (Vosk)	None, English only	LoRa	Not checked	Yes

System	Speech on device	Indian languages	Device-to-device link	Store-and-forward	Works below 1 kbps
Briar	No	Not applicable	Bluetooth, Wi-Fi, Tor	Yes	No
Codec2, Lyra	Codec only	Not applicable	No	No	Codec2 700C, marginal
Other SIH26173 repos	Yes	2 to 10 claimed	Bluetooth, Wi-Fi Direct	Rare	Emulated in at least one

Gap register. Grades: A documented, B strongly supported engineering gap, C reasonable inference, D unsupported assumption.

Gap	Evidence	Grade
Flagship offline translation covers one Indian language on one phone family	Android Authority, BGR	A
Best Indian ASR is too big for low-end phones	600M model needs about 2.5 GB RAM even in int8 (build card); one team audit found a 2.18 GB compulsory bundle	A
Models lose accuracy on spontaneous speech	BhasaAnuvaad reports read-versus-spontaneous gap	A
No public on-phone accuracy, speed and RAM numbers for Indic ASR plus TTS	Nothing found; team repos give a few self-measured figures	B
Licence-clean TTS for all ten languages is unproven	MMS is CC-BY-NC; voice licences vary; team reports conflict on Gujarati, Kannada, Odia	B
UTF-8 wastes 2.4–2.8× on Indic text	My measurement above; other repos use zlib, Protobuf or codebooks, and none I read handles script size explicitly	B
Demos run on wide links and cannot show the narrow-link benefit	Only one repo I read has a throttle test	B

Gap	Evidence	Grade
Recognition errors reach the listener as fluent wrong speech, with no uncertainty signal	No public implementation found that transmits confidence; my search was limited	C
Field devices need internet once to fetch models	tanmayjoshi01 notes provisioning needs the internet; peer-to-peer model sharing is an unclaimed fix	C
Code-switched Hindi-English speech and regional accents	Known research problem, not solvable by a student build	B, do not claim
Wi-Fi Direct and Bluetooth behave differently across phone makers	Reported by several team repos	B
"Nobody converts speech to text for narrow links"	Contradicted by STCTS and the Meshtastic ATAK plugin	D
"Existing apps do not work offline"	Contradicted by Pixel offline Live Translate, Briar, Meshtastic	D

The smallest real gap worth solving. No public system I found combines all of the following for Indian languages, on ordinary phones, with numbers behind it: on-device recognition and speech output, an Indic-efficient message format, a way to convey recognition uncertainty to the listener, reliable delivery over a link that is slow, lossy and intermittent, and benchmarks that show the trade-offs honestly.

8. Candidate directions and the proposed solution

Combine the required baseline with three measured contributions (efficient Indic wire format, uncertainty-aware speech path, reliable delivery under an emulated narrow link); the directions that depend on multi-hop mesh, LoRa hardware or full translation are stretch goals, not the core.

Directions compared

Direction	Already exists	Gap left	Effort	Fit for a student demo	Novelty	Main risk, and the jury question
1. Communication-aware multilingual voice (adapts to link quality)	Adaptive codecs such as Opus	Applying it to text-mediated speech	Medium	Good with an emulator	Medium	Unclear metric. "What exactly adapts?"
2. Offline-first speech communication	Many SIH repos, Meshtastic ATAK plugin	Measured Indic accuracy, speed, RAM	Medium	Required baseline	Low	"How is this different from the other teams?"
3. Store-and-forward multilingual voice	Briar, Airpass for text; none found for speech	Speech outbox with TTL and priority	Medium	Green for outbox and ACK, yellow for multi-hop	Medium	The PS may only require Wi-Fi/Bluetooth, so mesh could be off-rubric
4. Extremely low-bandwidth text-mediated voice	STCTS, ATAK plugin	Script-aware Indic coding, narrow-link proof	Low	Green	Low to medium	"How much do you really save over Codec2?" Answer: 73–87%
5. On-device Indic stack	Model cards, sherpa-onnx	A tested ten-language pack system with licences	High	Yellow (TTS coverage)	Medium	"Does all of it run on a cheap phone?"
6. Language-adaptive (automatic language ID)	Language-ID models exist	Short-utterance accuracy is unknown to me	Medium	Yellow; keep manual choice as default	Low	Wrong pack chosen silently
7. Optimised for intermittent connectivity	DTN literature, Briar	Speech-specific behaviour, reconnect timing	Medium	Green with emulator	Medium	Needs a clear demo scenario
8. Uncertainty-aware speech path (my addition)	Not found publicly for this PS	Convey recognition confidence to sender and listener	Medium	Green	Medium to high	"How do you compute confidence, and is it calibrated?"

Proposed solution: a verified semantic voice link

The baseline (direction 2) plus directions 4, 7 and 8. Three contributions, each testable:

1. **Indic-efficient wire format.** One byte per Indic character, a 12-byte header, an authenticated tag, and an optional shared phrase codebook for alerts. Target: at least 2.4× smaller than UTF-8 on real transcripts (the toy-corpus result to reproduce).
2. **Uncertainty-aware speech path.** The sender hears the recognized text read back and can accept or redo. Each word carries one confidence bit. A small lexicon of critical tokens (numbers, place words, medical terms) forces confirmation when confidence is low. The receiver's speech engine marks unclear words. Measure how many wrong sentences reach a listener with and without this.
3. **Reliable delivery under an emulated narrow link.** ACK, retry with backoff, de-duplication, ordering, priority, and an outbox with a time-to-live, exercised by a link emulator sweeping rate, loss, jitter and disconnects.

Why this is defensible. Each claim is a number you measure against an obvious baseline (UTF-8 text, no confidence signal, no retry), not a feature list. It sits inside the PS's evaluation criteria: accuracy, latency and efficiency.

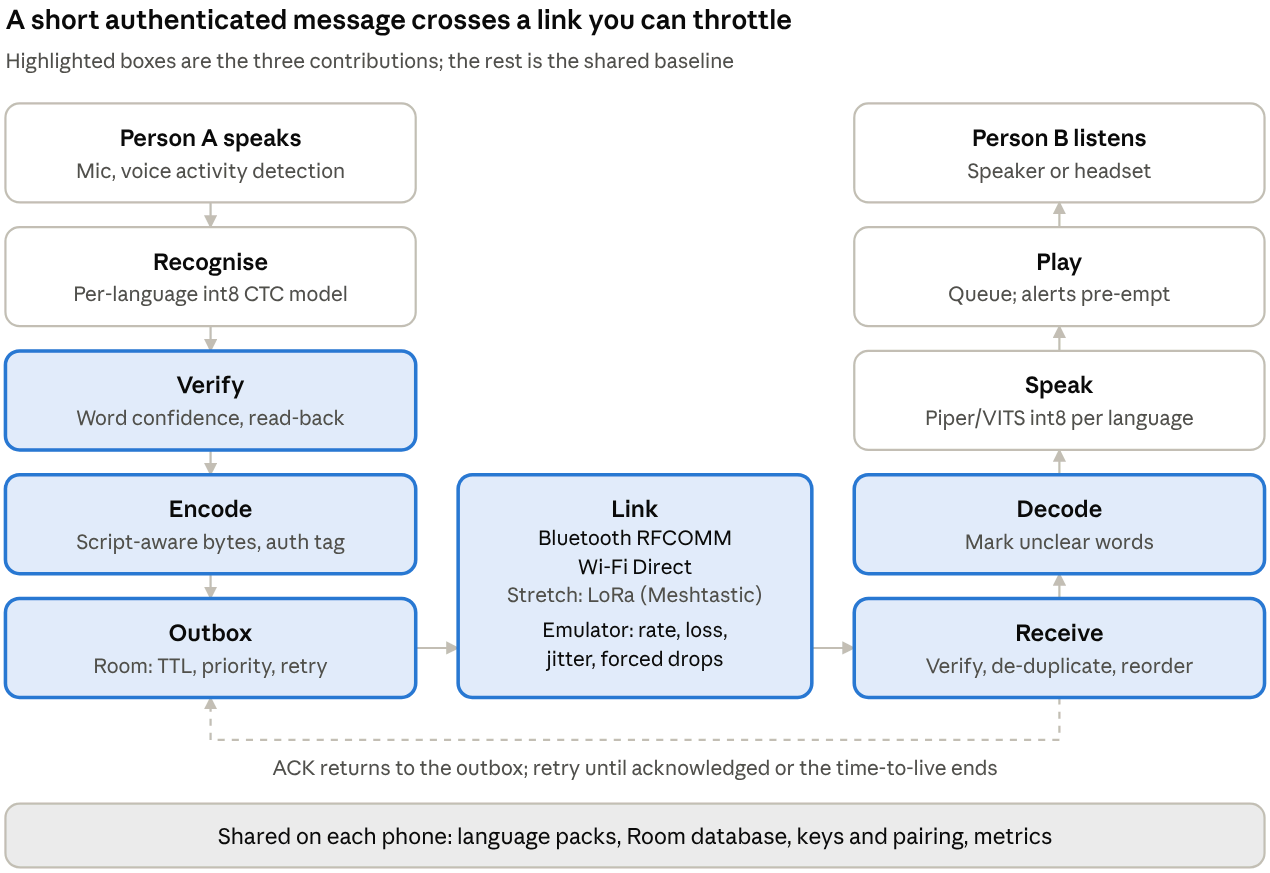
Why it is feasible. Every component is open source and already run on phones by other teams; the new work is a wire format, a confidence signal, an outbox and an emulator, all of which can be built and tested without special hardware.

What is not novel, and you should say so: speech to text to speech over Bluetooth or Wi-Fi Direct, Protobuf packets, phrase codebooks (another team has one), sherpa-onnx pipelines.

Open assumption to confirm: the confidence signal depends on getting per-token probabilities from the recognizer. Check in week one whether sherpa-onnx exposes them for the CTC model, or whether you need your own ONNX Runtime decoder (one team already wrote one).

9. Complete architecture

Two identical phones run the same stack: a phrase is recognised, checked, encoded and queued on one, crosses any transport, then is verified, decoded and spoken on the other. No backend is needed.



architecture · sender down the left, receiver up the right

Read it as a U: the sender's steps run down the left, the message crosses the link at the bottom, the receiver's steps run up the right, and the ACK returns along the dashed line.

Layer	Choice	Why
1. Interface	Kotlin with Jetpack Compose, one foreground service owning audio and network	Direct control of AudioRecord, Bluetooth and Wi-Fi Direct sockets, and background limits; one team used Flutter, but native avoids plugin surprises in audio and sockets
2. Voice processing	16 kHz mono capture, voice-activity detection (Silero, under 1.5 MB per one team), optional noise suppression	Stops the recognizer running on silence, which protects battery and latency

Layer	Choice	Why
3. Language intelligence	Per-language recognition pack, text normalisation (Unicode NFC, digits, danda), language chosen manually by default	Automatic language ID on short clips is unproven, so keep it a stretch
4. Local data	Room over SQLite	Outbox, de-duplication window, peers, language packs, metrics
5. Communication manager	One transport interface with Bluetooth RFCOMM, Wi-Fi Direct, Nearby fallback, and a link emulator	Keeps speech code independent of the radio
6. Messaging protocol	Compact binary format, ACK, retry, de-duplication, priority	Turns an unreliable link into delivery you can measure
7. Speech output	Per-language Piper/VITS int8, system TTS as fallback	Small, offline, licence recorded per voice
8. Security	Paired identity keys, authenticated encryption, replay window	Voice is personal data; alerts must be trustworthy
9. Backend	None	Peer-to-peer meets the PS; a server contradicts the offline goal
10. Monitoring	In-app diagnostics screen and CSV export; laptop scripts for error-rate sweeps	Evidence for the jury without infrastructure

Communication architecture

The message is about 75 bytes and the link is the least reliable part, so the protocol's job is to make delivery measurable, not to save more bytes.

Discovery and connection. Peers pair once and then reconnect by identity.

- **Bluetooth:** pair the first time with a QR code or short code, then find known peers from the paired list and open an RFCOMM socket on an app-specific service UUID.
- **Wi-Fi Direct:** advertise the app's service, discover it on the other phone, negotiate the group owner, then open a TCP socket on a fixed port.

- **Nearby (fallback):** advertise and discover with a shared service ID under the P2P_CLUSTER strategy.
- **Link choice:** the transport manager keeps the best live link per peer (Wi-Fi Direct for range, Bluetooth as fallback), sends a heartbeat, and reconnects with backoff.

Wire format (proposal, 12-byte header)

Field	Bytes	Meaning
Version and type	1	4-bit version; type: text, phrase, alert, ACK, hello, pack offer
Flags and priority	1	2-bit priority (routine, urgent, alert); needs-ACK, has-confidence, fragment, codebook
Language	1	Index into the language registry
Sender id	2	Short id assigned at pairing
Sequence	2	Per-sender counter
Time	2	Minutes since a session epoch, for time-to-live and ordering
Hops	1	0 when direct; used only if relay is enabled
Length	1	Payload bytes
Reserved	1	Future use
Payload	n	Script-aware text bytes, then one confidence bit per word
Authentication tag	16	Nonce derived from session key, sender id and sequence; not sent

A decoded message reads like `{sender: 0x12A4, seq: 1024, lang: mr, type: TEXT, priority: routine, text: "…", conf: 1101…}`. Fragmentation exists only for narrow links such as LoRa, where a payload above about 237 bytes must be split.

Reliability

Event	Behaviour
Packet lost	No ACK arrives; retry with exponential backoff and jitter (suggested start: 1 s, doubling to a 30 s cap) until the time-to-live ends
Device disconnects mid-send	Message returns to queued; resend whole on reconnect (messages are tiny, so no partial resume)
Receiver unavailable	Outbox holds it until expiry (suggested: routine 10 min, alert 60 min; tune with the jury scenario)
Link vanishes halfway	Same as a disconnect; the receiver discards partial frames
Duplicate arrives	Dropped using a window of recent (sender, sequence) pairs, stored in Room
Out of order	Play in sequence; if a gap lasts about 5 s, play the later message with a "previous message missing" cue
Alert	Highest priority: sent first, skips the ordering wait, pre-empts playback

Store-and-forward is the outbox: a Room table of messages with state (queued, sending, sent, acknowledged, expired), retry counter, next-retry time and expiry. A link-up event flushes it by priority and then age. Multi-hop relay (flood with a hop limit plus de-duplication, as in Airpass) is a stretch goal and probably off-rubric.

Offline-first behaviour

Situation	What the app does
Fully online	Nothing changes; the message path never uses the internet. Internet is only for downloading language packs and updates
Poor connectivity	Same path with more retries; the screen shows per-message state and link quality

Situation	What the app does
Completely offline	Normal operation, provided the language packs are already installed or shared from a peer
Receiver offline	Message waits in the outbox; sender sees "waiting" and can cancel
Connection returns	Outbox flushes by priority then age; receiver de-duplicates; routine messages older than a staleness limit are shown as text, not spoken
Several devices reachable	One conversation and queue per peer; alerts go to every reachable peer; group send is a stretch